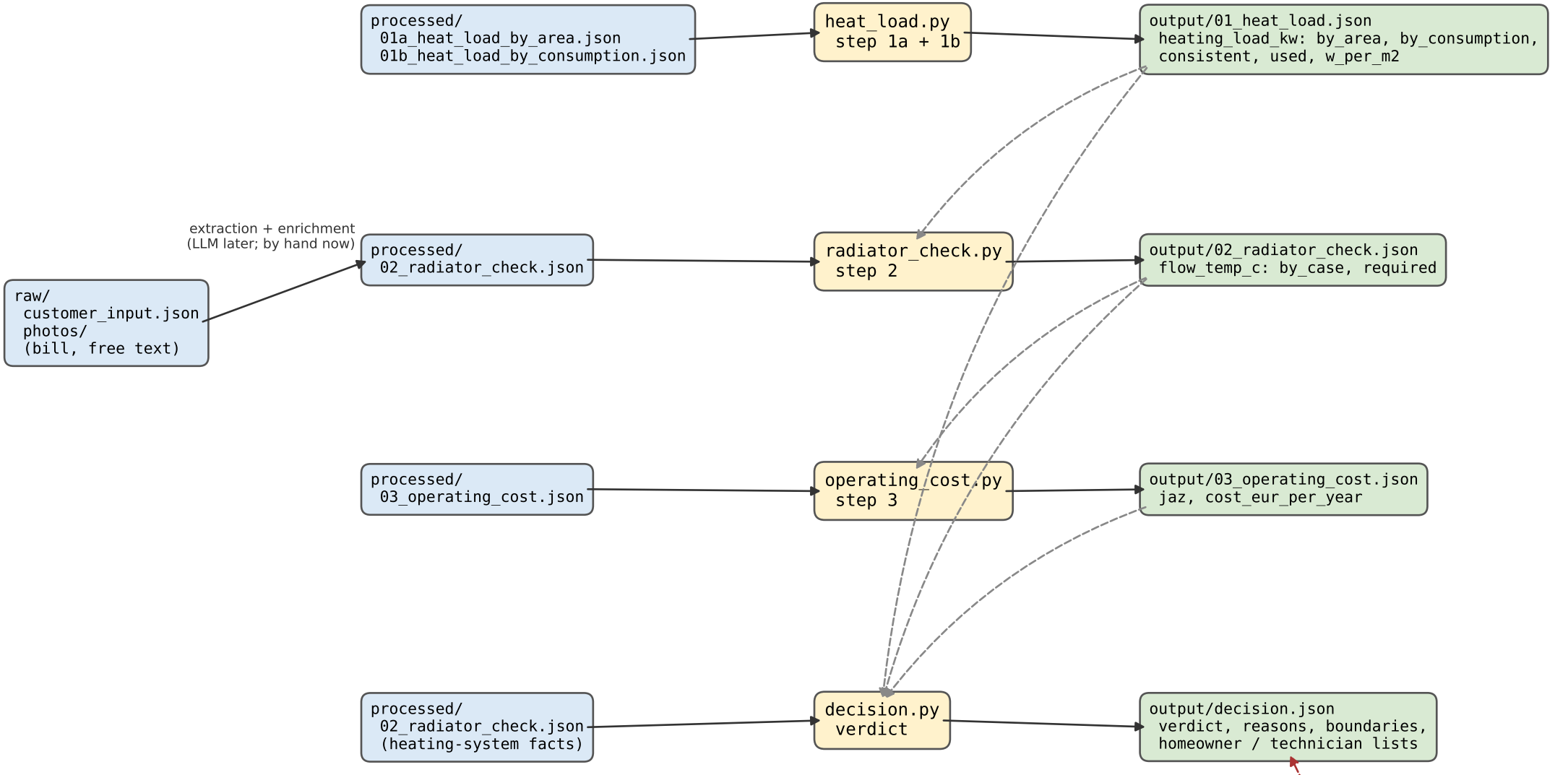
Decision engine: data flow

data/demo/<customer>/raw -> processed -> output. Python, stdlib only. Run: python3 scripts/run_engine.py <customer>



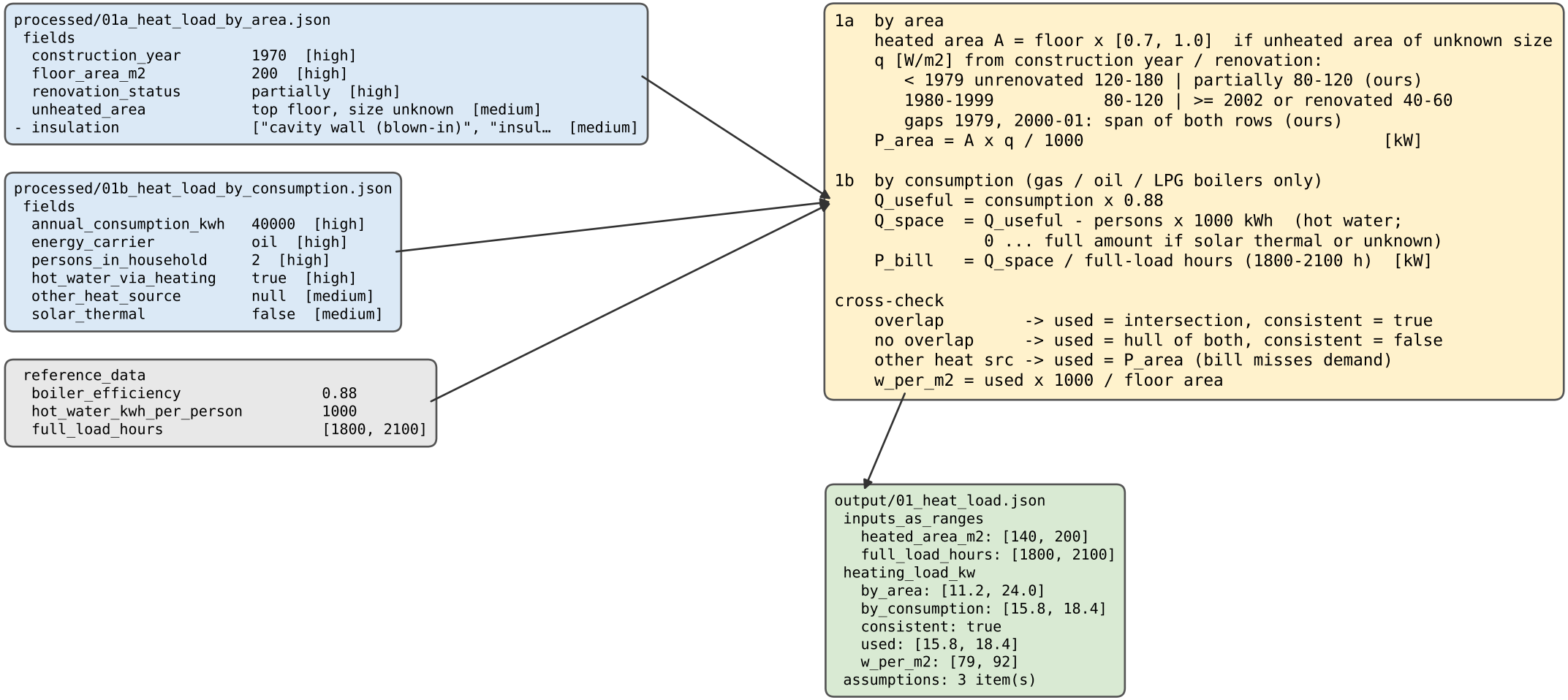
Legend
 blue   input files
 yellow engine step (src/decision_engine/)
 green output files
 dashed grey: output read by a later step

Conventions: ranges [min, max], unknown = null,
every field {value, source, confidence, evidence}

run_engine.py adds dev_check to decision.json;
the engine itself never sees the labels.

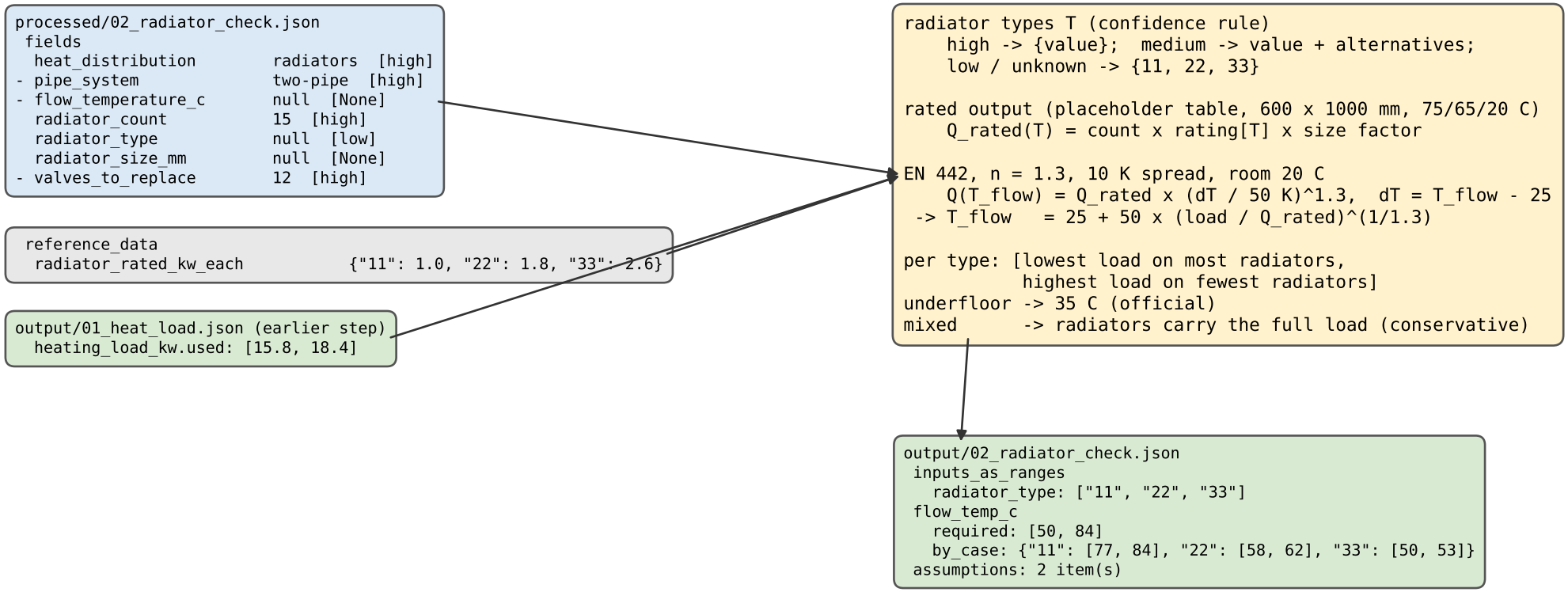
Step 1: heat load (heat_load.py)

Values shown for B-03 (data/demo/B-03). '-' in front of a field: carried in the file but not used by this step.



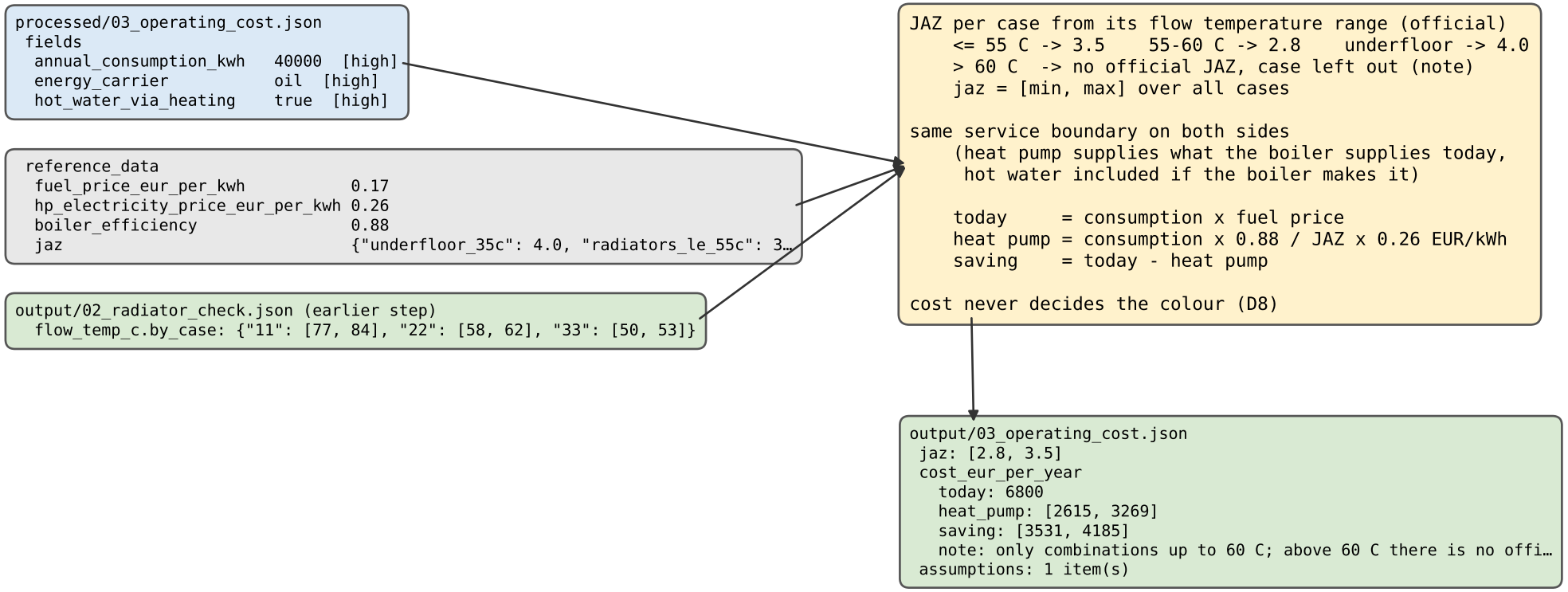
Step 2: radiator check (radiator_check.py)

Values shown for B-03 (data/demo/B-03). '-' in front of a field: carried in the file but not used by this step.



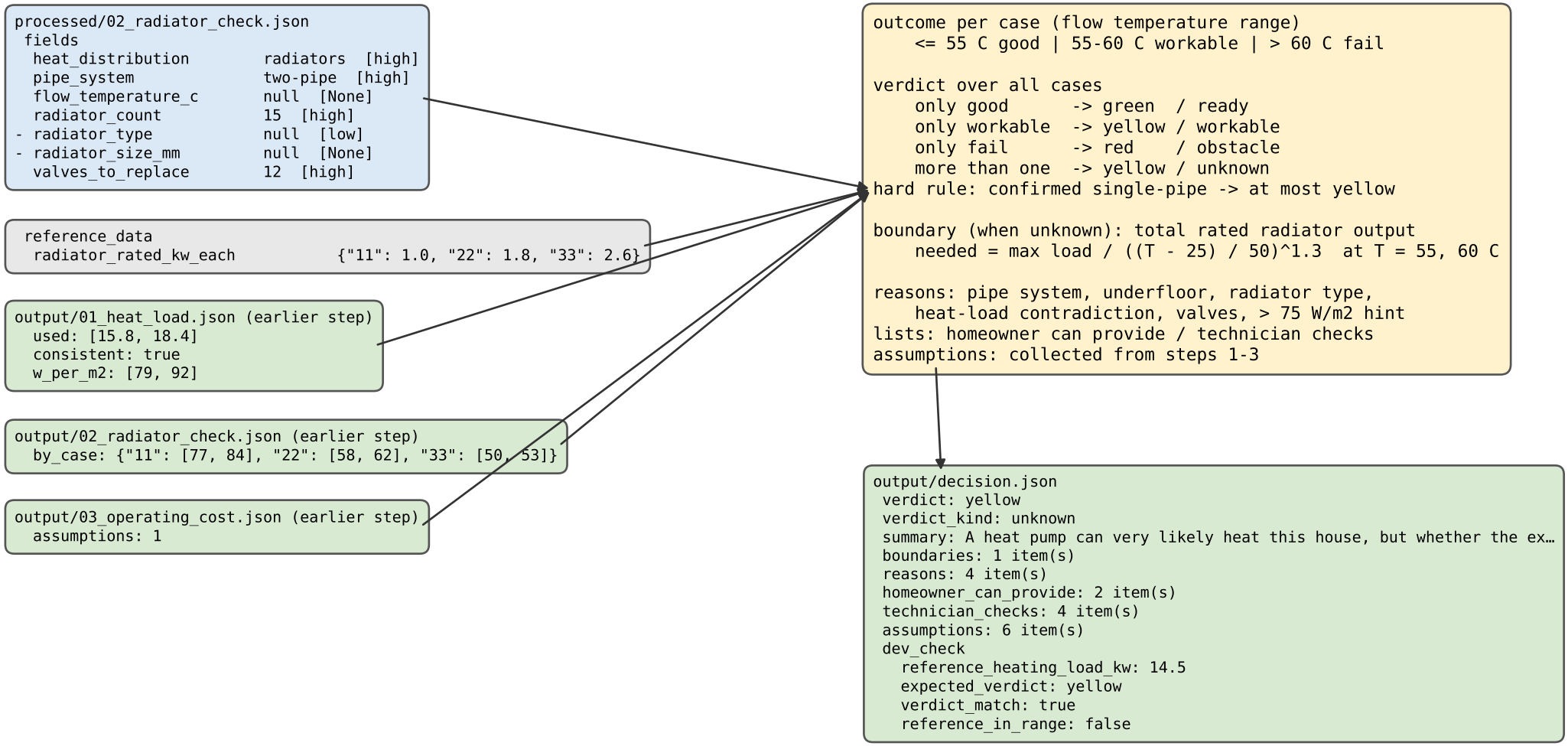
Step 3: operating cost (operating_cost.py)

Values shown for B-03 (data/demo/B-03). '-' in front of a field: carried in the file but not used by this step.



Step 4: decision (decision.py)

Values shown for B-03 (data/demo/B-03). '-' in front of a field: carried in the file but not used by this step.



Results: all demo customers

Engine output per customer; dev labels from data/raw/buildings/buildings_dev.json (B-01 to B-03 only).

	verdict	load used kW	consistent flow C by radiator type	today -> HP EUR/a	expected	ref kW in range
B-01	yellow/unknown	[2.3, 23.4]	False	11:37-99 22:33-72 33:31-61	None -> [654, 817]	green MISS 9.6	True
B-02	green/ready	[7.2, 10.8]	None	underfloor:35-35	1705 -> [887, 887]	green OK 9.7	True
B-03	yellow/unknown	[15.8, 18.4]	True	11:77-84 22:58-62 33:50-53	6800 -> [2615, 3269]	yellow OK 14.5	False
B-04	yellow/unknown	[9.0, 19.2]	False	11:65-97 22:50-71 33:44-59	4250 -> [1634, 2043]	-	-	-
B-05	yellow/unknown	[5.0, 13.2]	False	11:60-98 22:47-72 33:42-60	2970 -> [1765, 2206]	-	-	-

Dev check: verdict 2/3 match, reference heat load inside 'used' 2/3

